

Formula for getting a square diagonal

$$D1 = s * \sqrt{2}$$

s, above is side of square, and side of rectangle

calculus

$$D1 = 2 * \sqrt{2}$$

using sin

$$\sin(45) = \frac{s}{D1} \quad (1)$$

$$\sin(60) = \frac{s}{D2} \quad (2)$$

multiplication D1 by both sides on (1)